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Aburub et al.

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(54) **PROCESSES AND INTERMEDIATE FOR THE LARGE-SCALE PREPARATION OF 2,4,6-TRIFLUORO-N-[6-(1-METHYL-PIPERIDINE-4-CARBONYL)-PYRIDIN-2-YL]-BENZAMIDE HEMISUCCINATE, AND PREPARATION OF 2,4,6-TRIFLUORO-N-[6-(1-METHYL-PIPERIDINE-4-CARBONYL)-PYRIDIN-2-YL]-BENZAMIDE ACETATE**

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CPC **C07D 401/06** (2013.01); **A61K 9/2095** (2013.01); **C07C 51/412** (2013.01); **C07B 2200/13** (2013.01)

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CPC .. C07D 401/06; A61K 9/2095; C07C 51/412; C07B 2200/13

See application file for complete search history.

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(57) **ABSTRACT**

The embodiments of present invention provide processes for the large-scale preparation of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hemisuccinate, and formulations and product forms, such as tablets, made by these processes. Additionally, embodiments of the present invention provide tablets including 25 mg, 50 mg, 100 mg, or 200 mg free base equivalent of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-pyridin-2-yl]-benzamide hemisuccinate.

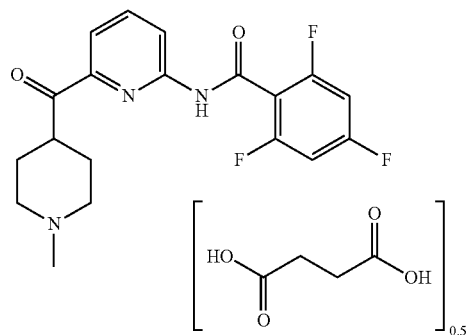
4 Claims, 1 Drawing Sheet

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**PROCESSES AND INTERMEDIATE FOR
THE LARGE-SCALE PREPARATION OF
2,4,6-TRIFLUORO-N-[6-(1-METHYL-
PIPERIDINE-4-CARBONYL)-PYRIDIN-2-YL]-
BENZAMIDE HEMISUCCINATE, AND
PREPARATION OF
2,4,6-TRIFLUORO-N-[6-(1-METHYL-
PIPERIDINE-4-CARBONYL)-PYRIDIN-2-YL]-
BENZAMIDE ACETATE**

The embodiments of the present invention relate to the fields of pharmaceutical chemistry and synthetic organic chemistry, and provide processes and an intermediate for the large-scale synthesis of 2,4,6-trifluoro-N-[6-(1-methyl-piperidine-4-carbonyl)-pyridin-2-yl]-benzamide hemi-succinate salt, a 5-HT_{1F} receptor agonist, and formulations and product forms made by these processes, and to preparation of 2,4,6-trifluoro-N-[6-(1-methyl-piperidine-4-carbonyl)-pyridin-2-yl]-benzamide acetate and uses thereof for parenteral formulations and treatment of migraine.

Lasmiditan is a selective and highly potent 5-HT_{1F} receptor agonist which is now approved in the United States, as 50 mg or 100 mg tablets, for acute on-demand treatment of migraine (See e.g. Rubio-Beltrán et al., *Pharmacol Ther* 2018; 186:88-97, and *Lasmiditan for the Treatment of Migraine*, Capi, M. et al., Expert Opinion Investigational Drugs, (2017), Vol. 26, NO. 2, 227-234). Lasmiditan (COL 144, LY 573144, CAS Registry No. 439239-90-4) can be described chemically as 2,4,6-trifluoro-N-[6-(1-methyl-piperidin-4-ylcarbonyl)-pyridin-2-yl]-benzamide. U.S. Pat. No. 7,423,050 and U.S. Publication No. 20080300407 describe the hemisuccinate salt of 2,4,6-trifluoro-N-[6-(1-methyl-piperidine-4-carbonyl)-pyridin-2-yl]-benzamide having the structural formula:



Methods of preparing lasmiditan and salts and certain polymorphic forms, formulations, and dosage forms thereof, are known to the skilled artisan, and are described for example in WO 2003/084949, WO 2011/123654, and WO 2018/106657.

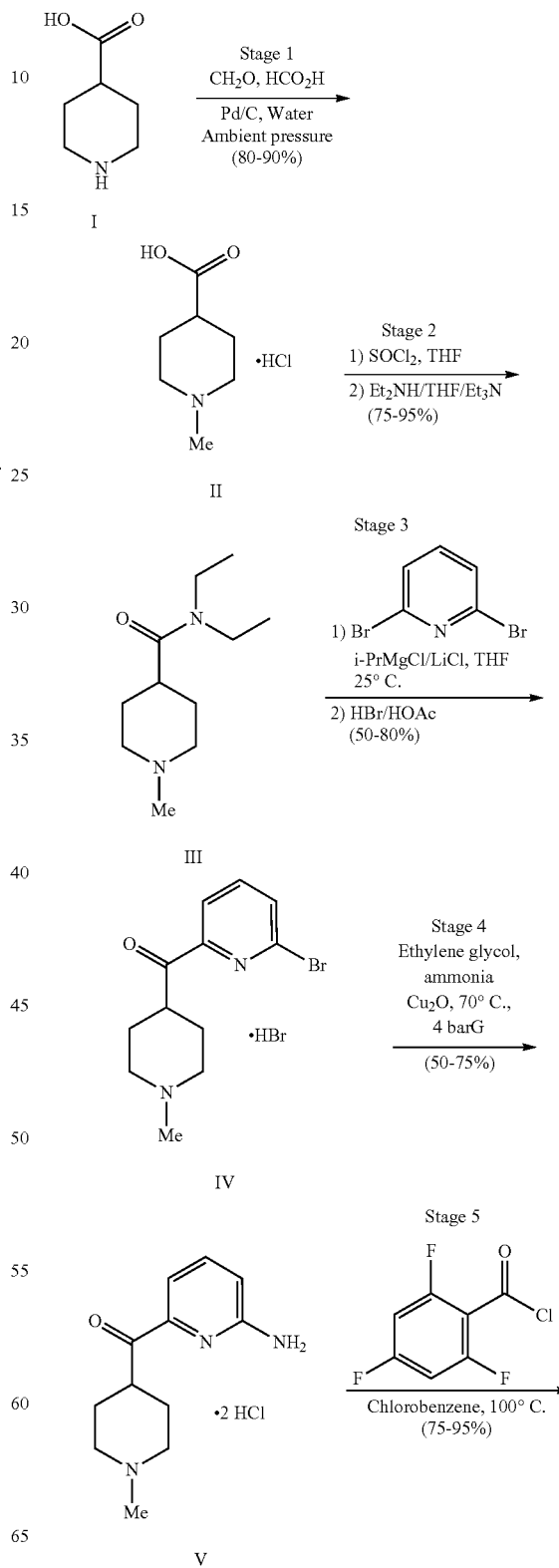
As used herein, useful forms of lasmiditan include pharmaceutically acceptable salts thereof, including but not limited to 2,4,6-trifluoro-N-[6-(1-methyl-piperidin-4-ylcarbonyl)-pyridin-2-yl]-benzamide mono-hydrochloride salt, and 2,4,6-trifluoro-N-[6-(1-methyl-piperidine-4-carbonyl)-pyridin-2-yl]-benzamide hemi-succinate salt. A synthetic route for the preparation of the hemi-succinate salt of 2,4,6-trifluoro-N-[6-(1-methyl-piperidine-4-carbonyl)-pyridin-2-yl]-benzamide has been disclosed previously as shown below in Scheme A. The overall yield of lasmiditan starting with commercially available piperidine 4-carboxylic acid via the route described in Scheme A below is about

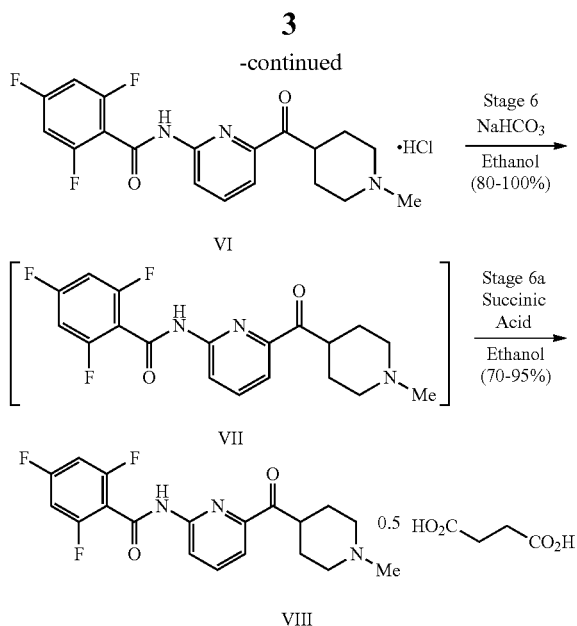
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10-46% over all 9 steps. Improvements in the synthesis of lasmiditan could provide substantial and varied benefits, particularly for production at large-scale.

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Scheme A





Synthetic chemistry process routes may be redesigned or revised aiming to achieve various advantages, including for example: improved yields, obtaining crystalline products, decreasing impurity profiles, utilizing commercially available intermediates, minimizing the number of synthetic steps needed, reducing the inputs required and/or the by-products produced, or any useful combination of such improvements, to achieve important real-world outcomes including decreased costs, providing less resource intensive processes, and facilitating efficient production. Improved methods of making lasmiditan are needed which may achieve one or more of these aims, particularly for large-scale synthesis.

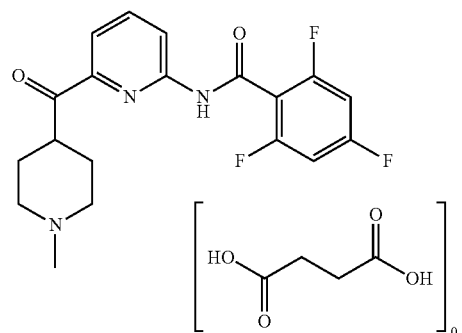
Further, migraine is one of the most common presenting symptoms in emergency rooms. Current methods for headache relief in the emergency room setting, when using lasmiditan for patients who have difficulty administering a tablet due to nausea and/or vomiting, may need to rely on the preparation of a diluted formulation of about 1 mg/ml lasmiditan delivered intravenously over an extended period of time, for example from about 20-60 minutes. Lasmiditan has been delivered intravenously in clinical studies in doses from about 1-60 mg delivered in 60 ml infusions over 20 minutes (See US Patent Application Publication No. 2010/0256187). The safe and effective treatment of migraine with lasmiditan for patients unable to administer tablets would be enabled by the availability of a high concentration parenteral dosage form. The present disclosure also addresses this need.

SUMMARY

The embodiments of the present invention provide processes for the preparation of lasmiditan hemisuccinate, 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hemisuccinate salt, and/or compositions thereof, and/or particularly useful intermediates for use in these processes. The embodiments of the present invention further provide for the preparation of lasmiditan acetate, 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide acetate salt, and/or compositions thereof, and/or uses of lasmiditan acetate, and formulations thereof, in subcutaneous drug delivery.

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In one embodiment, referred to as Route I, the present invention provides a process for preparing a compound of the formula:



comprising the steps of:

- i.) Treatment of piperidine-4-carboxylic acid under reductive amination conditions comprising formaldehyde and formic acid in water with subsequent treatment with aqueous HCl followed by water distillation and acetonitrile addition, with repeated dilution/distillation until the water content is not more than 0.2% by Karl-Fischer analysis, to obtain solid 1-methylpiperidine-4-carboxylic acid hydrochloride;
- ii.) Treatment of 1-methylpiperidine-4-carboxylic acid hydrochloride with a chlorinating agent such as thionyl chloride in chlorobenzene obtain 1-methylpiperidine-4-carboxylic acid chloride;
- iii.) Treatment of 1-methylpiperidine-4-carboxylic acid chloride with N,N-diethylamine in chlorobenzene containing triethylamine with subsequent base wash and subsequent treatment with aqueous HCl in isopropanol to obtain solid N,N-diethyl-1-methyl-piperidine-4-carboxamide hydrate hydrochloride;
- iv.) Treatment of N,N-diethyl-1-methyl-piperidine-4-carboxamide hydrate hydrochloride with a mineral base such as aqueous NaOH in a non-polar solvent such as methyl-tert-butyl ether with subsequent water wash, phase separation, and distillation of the organic solvent until the water content is not more than 0.1 weight % by Karl Fischer analysis to obtain N,N-diethyl-1-methyl-piperidine-4-carboxamide;
- v.) Subsequent treatment of N,N-diethyl-1-methyl-piperidine-4-carboxamide with (6-bromo-2-pyridyl)lithium in a non-polar organic solvent such as methyl-tert-butyl ether with subsequent extraction of the resulting mixture with water and a suitable organic solvent such as n-butanol, phase separation, and repeated distillation of the organic solvent until the water content is not more than 0.2 weight % by Karl-Fischer analysis, to obtain (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone;
- vi.) Treatment of (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone with aqueous HBr and subsequent extraction with n-butanol followed by repeated distillation of the organic solvent until the water content is not more than 0.3% by Karl-Fischer analysis, to obtain solid (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone hydrobromide;
- vii.) Treatment of (6-bromo-2-pyridyl-1-methyl-4-piperidyl)methanone hydrobromide with a solution of NH_3 in ethylene glycol in the presence of Cu_2O catalyst at about 80° C. for about 2 hr, with subsequent washes

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with water, saturated aqueous NaCl, and 20% aqueous NaOH and subsequent extraction with a non-polar aprotic solvent such as methyl-tert-butyl ether, phase separation, and treatment of the organic phase with 5 weight % carbon;

viii.) Filtration of the above mixture, dilution with a suitable polar alcoholic solvent such as isopropanol, and repeated distillation of the organic solvent until the water content is not more than 0.2% by Karl-Fischer analysis, with subsequent treatment of the resulting residue with isopropanol, water, and 20 weight % HCl, wherein the water concentration of the resulting slurry is at least 2%, filtration of the resulting slurry, and drying under vacuum at 40° C. for 16-24 hr to obtain (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride;

ix.) Treatment of (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride in chlorobenzene with 6 weight/weight % NaOH in water at about 54° C. for about 30 min, with subsequent phase separation and vacuum distillation of the aqueous solution to obtain (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone;

x.) Subsequent treatment of (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone with 2,4,6-trifluorobenzoic acid chloride in chlorobenzene at about 100° C. for about 4 hr, with subsequent cooling, charging with acetonitrile and heating the resulting slurry to 80° C. for about 1 hr, and subsequent collection of the resulting solid by filtration, to obtain solid 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hydrochloride;

xi.) Treatment of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hydrochloride with saturated aqueous Na₂CO₃ in methyl-tert-butyl ether;

xii.) Treatment of the mixture of step xi above with SiO₂ with subsequent filtration, treatment with carbon, filtration, and evaporation, dilution with ethanol, and distillation until the water content is not more than 1% by Karl-Fischer analysis, to obtain 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide;

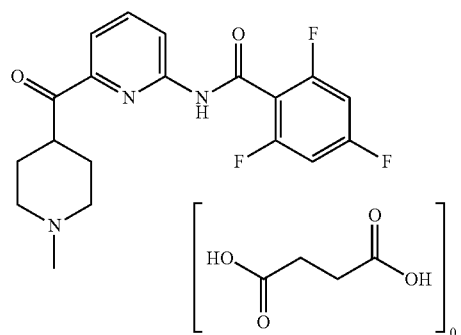
xiii.) Treatment of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide in ethanol with a solution of 0.5 equivalents succinic acid in ethanol at about 55° C. for not less than 3 hr at RT, and subsequent collection of the solid by filtration, to obtain solid 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hemisuccinate.

In the process of Route I above, preferably the reactions are performed using batch processing methodology. In an embodiment the batches by Route I are produced at process scale. In an embodiment the batches by Route I are produced in at least 1 kilogram. In an embodiment the batches by Route I are produced in at least 10 kilograms. In an embodiment the batches by Route I are produced in at least 100 kilograms.

In the process of Route I above, the use of chlorobenzene avoids degradation which occurs under alternative methods, such as THF, which reacts with the acid chloride under scale (e.g., 100 kg) resulting in essentially no yield of the acid chloride.

In another embodiment, referred to as Route II, the present invention provides a process for preparing a compound of the formula:

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comprising the steps of:

i.) Treatment of piperidine-4-carboxylic acid under reductive amination conditions comprising formaldehyde and formic acid in water with subsequent treatment with aqueous HCl followed by water distillation and acetonitrile addition, with repeated dilution/distillation until the water content is not more than 0.2% by Karl-Fischer analysis, to obtain solid 1-methylpiperidine-4-carboxylic acid hydrochloride;

ii.) Treatment of 1-methylpiperidine-4-carboxylic acid hydrochloride with a chlorinating agent such as thionyl chloride in chlorobenzene to obtain 1-methylpiperidine-4-carboxylic acid chloride;

iii.) Treatment of 1-methylpiperidine-4-carboxylic acid chloride with N,N-diethylamine in chlorobenzene containing triethylamine with subsequent base wash and subsequent treatment with aqueous HCl in isopropanol to obtain solid N,N-diethyl-1-methyl-piperidine-4-carboxamide hydrate hydrochloride;

iv.) Treatment of N,N-diethyl-1-methyl-piperidine-4-carboxamide hydrate hydrochloride with a mineral base such as aqueous NaOH in a non-polar solvent such as methyl-tert-butyl ether with subsequent water wash, phase separation, and distillation of the organic solvent until the water content is not more than 0.1 weight % by Karl Fischer analysis to obtain N,N-diethyl-1-methyl-piperidine-4-carboxamide;

v.) Subsequent treatment of N,N-diethyl-1-methyl-piperidine-4-carboxamide with (6-bromo-2-pyridyl)lithium in a non-polar organic solvent such as methyl-tert-butyl ether with subsequent extraction of the resulting mixture with water and a suitable organic solvent such as n-butanol, phase separation, and repeated distillation of the organic solvent until the water content is not more than 0.2 weight % by Karl-Fischer analysis, to obtain (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone;

vi.) Treatment of (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone with aqueous HBr and subsequent extraction with n-butanol followed by repeated distillation of the organic solvent until the water content is not more than 0.3% by Karl-Fischer analysis, to obtain solid (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone hydrobromide;

vii.) Treatment of (6-bromo-2-pyridyl-1-methyl-4-piperidyl)methanone hydrobromide in a biphasic mixture of water and toluene with solid KOH for about 3 hr with subsequent separation of the organic layer and evaporation of the solvent to obtain of (6-bromo-2-pyridyl-1-methyl-4-piperidyl)methanone;

viii.) Treatment of (6-bromo-2-pyridyl-1-methyl-4-piperidyl)methanone with 2,4,6-trifluorobenzamide in

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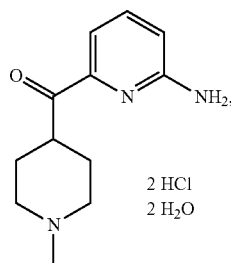
toluene containing K_2CO_3 , water, $Pd(OAc)_2$, and Xantphos at about 70° C. for about 12 hr, until the (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone content is not more than 0.1% by HPLC, with subsequent dilution of the reaction mixture with water and EtOAc, subsequent treatment with thiourea-modified silica gel at 60° C. for about 8 hr, with subsequent filtration to obtain a solution of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide;

ix.) Treatment of a solution of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide in EtOAc with a solution of about 0.5 equivalents of succinic acid dissolved in EtOH at 55° C. for about 3 hr, with subsequent cooling to RT over about 10 hr, and collection of the resulting solids by filtration, to obtain solid 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hemisuccinate.

In the process of Route II above, preferably the reactions are performed using batch processing methodology. In an embodiment the batches by Route II are produced at process scale. In an embodiment the batches by Route II are produced in at least 1 kilogram. In an embodiment the batches by Route II are produced in at least 10 kilograms. In an embodiment the batches by Route II are produced in at least 100 kilograms.

In the process of Route II above, the use of chlorobenzene avoids degradation which occurs under alternative methods, such as THF, which reacts with the acid chloride under scale (e.g. 100 kg) resulting in essentially no yield of the acid chloride.

In another embodiment the present invention provides:

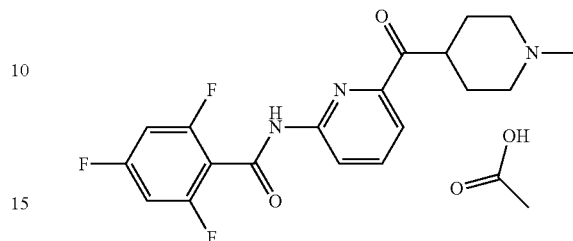


which can be named as (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride. Preferably this compound is crystalline. (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride is particularly useful in the preparation of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hemisuccinate, and processes which employ (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride may provide advantageous process characteristics, including but not limited to the purity of intermediate and/or final materials. (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride is believed to be a new stable hydrated form of 6-amino-2-pyridyl-(1-methyl-4-piperidyl)methanone. The process to isolate (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride described herein provides improved impurity rejection and an improved controlled crystallization process. Form and chemical stability studies showed (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride is generally stable, and drying studies show that over-drying to remove water of

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hydration is difficult, even under forcing conditions. Use of this intermediate provides advantageously high purity product at expected yield.

In another embodiment the present disclosure provides lasmiditan acetate, which can be represented by the formula:



In another embodiment the invention provides lasmiditan acetate in crystalline form, and further provides lasmiditan acetate in crystalline form characterized by an X-ray powder diffraction pattern using $CuK\alpha$ radiation having an intense peak at diffraction angle 2-theta of 26.2° in combination with one or more of the peaks selected from the group consisting of 20.4°, 14.0°, and 17.9° ($\pm 0.2^\circ$ respectively). In another embodiment the present invention provides a pharmaceutical composition comprising lasmiditan acetate according to the above embodiments with one or more pharmaceutically acceptable carriers, diluents, or excipients. Preferably the pharmaceutical composition comprises acetic acid. Preferably the pharmaceutical composition comprises acetic acid and is for subcutaneous administration.

In another embodiment the invention provides a method of treating migraine in a patient comprising administering to a patient in need of such treatment an effective amount of lasmiditan acetate. In another embodiment the invention provides a method of treating migraine in a patient comprising administering to a patient in need of such treatment an effective amount of lasmiditan acetate with one or more pharmaceutically acceptable carriers, diluents, or excipients. In another embodiment the invention provides a method of treating migraine in a patient comprising administering to a patient in need of such treatment an effective amount of lasmiditan acetate with one or more pharmaceutically acceptable carriers, diluents, or excipients wherein the composition comprises acetic acid.

In another embodiment the invention provides lasmiditan acetate for use in therapy. In another embodiment the invention provides a pharmaceutical composition of lasmiditan acetate with one or more pharmaceutically acceptable carriers, diluents, or excipients for use in therapy. In another embodiment the invention provides a pharmaceutical composition of lasmiditan acetate with one or more pharmaceutically acceptable carriers, diluents, or excipients, wherein the composition comprises acetic acid for use in therapy.

In another embodiment the invention provides lasmiditan acetate for use in the treatment of migraine. In another embodiment the invention provides a pharmaceutical composition of lasmiditan acetate with one or more pharmaceutically acceptable carriers, diluents, or excipients for use in the treatment of migraine. In another embodiment the invention provides a pharmaceutical composition of lasmiditan acetate with one or more pharmaceutically acceptable carriers, diluents, or excipients, wherein the composition comprises acetic acid for use in the treatment of migraine.

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BRIEF DESCRIPTION OF THE FIGURES

FIG. 1: Graphical representation of an ^1H NMR spectrum (400 MHz, DMSO-d_6) of lasmiditan acetate containing maleic acid (internal standard).

DETAILED DESCRIPTION

The reactions described herein may be performed via standard techniques known to the skilled artisan by employing routine glassware or may be performed on pilot and/or production scale in equipment designed for such transformations. Further, each of these reactions described may be executed via either a batch process, or where applicable, a flow reaction methodology. The term “batch process” as used herein refers to a process in which raw materials are combined in a reactor or vessel and product is removed at the end of the reaction.

Additionally, certain intermediates described in the following preparations may contain one or more nitrogen protecting groups. The variable protecting group may be the same or different in each occurrence depending on the particular reaction conditions and the particular transformations to be performed. The protection and deprotection conditions are well known to the skilled artisan and are described in the literature (See for example “*Greene’s Protective Groups in Organic Synthesis*”, Fourth Edition, by Peter G. M. Wuts and Theodora W. Greene, John Wiley and Sons, Inc. 2007).

The abbreviations listed below when used herein are defined as follows: “Å” means angstrom or angstroms. “ACN” means acetonitrile. “AcOH” means acetic acid. “Bn” means benzyl; “nBuLi” means n-butyllithium. “CAS No.” means Chemical Abstracts Registry number. “DCM” means dichloromethane. “DMF” means N,N-dimethylformamide. “DIPEA” means diisopropylethylamine. “DMSO” means dimethyl sulfoxide (perdeuterated $[\text{d}_6]$ if used for NMR). “EtOAc” means ethyl acetate. “EtOH” means ethanol or ethyl alcohol. “HBTU” means (2-(1H-benzotriazole-1-yl)-1,1,3,3-tetramethyluronium hexafluorophosphate. “HPLC” means high performance liquid chromatography. “HTRF” means homogeneous time-resolved fluorescence. “hr” or “h” means hour or hours. “IPA” means isopropyl alcohol. “IPC” means in-process control. “LAH” means lithium aluminum hydride. “LCMS” means liquid chromatography mass spectrometry. “LDA” means lithium diisopropylamide. “Me” as a substituent in a structural representation of a compound represents a methyl group. “MeOH” means methanol or methyl alcohol. “min” means minutes. “MS” means mass spectrometry or mass spectrum. “MTBE” means methyl tert-butyl ether. “NMR” means nuclear magnetic resonance. “NMT” means not more than. “OAc” means acetate. “psig” means pounds per square inch gauge. “PyBOP” means (benzotriazol-1-yl-oxytripyrrolidinophosphonium hexafluorophosphate). “RT” means room temperature/ambient temperature. “sec” means second or seconds as a unit of time. “TBS-Cl” means tert-butyldimethylsilyl chloride. “TEA” means triethylamine. “THF” means tetrahydrofuran. “tR” means retention time. “w/w” means weight to weight in a ratio.

Improved routes for the preparation of lasmiditan are provided below as Routes I and/or II, and other additional methods as provided below.

“Pharmaceutically acceptable salts” or “a pharmaceutically acceptable salt” refers to the relatively non-toxic, inorganic and organic salt or salts of the compounds of the present invention. It will be understood by the skilled artisan

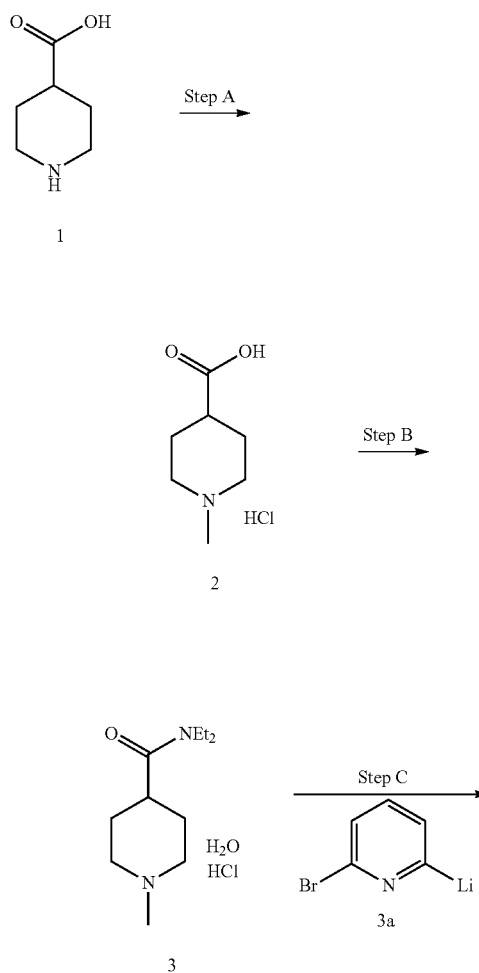
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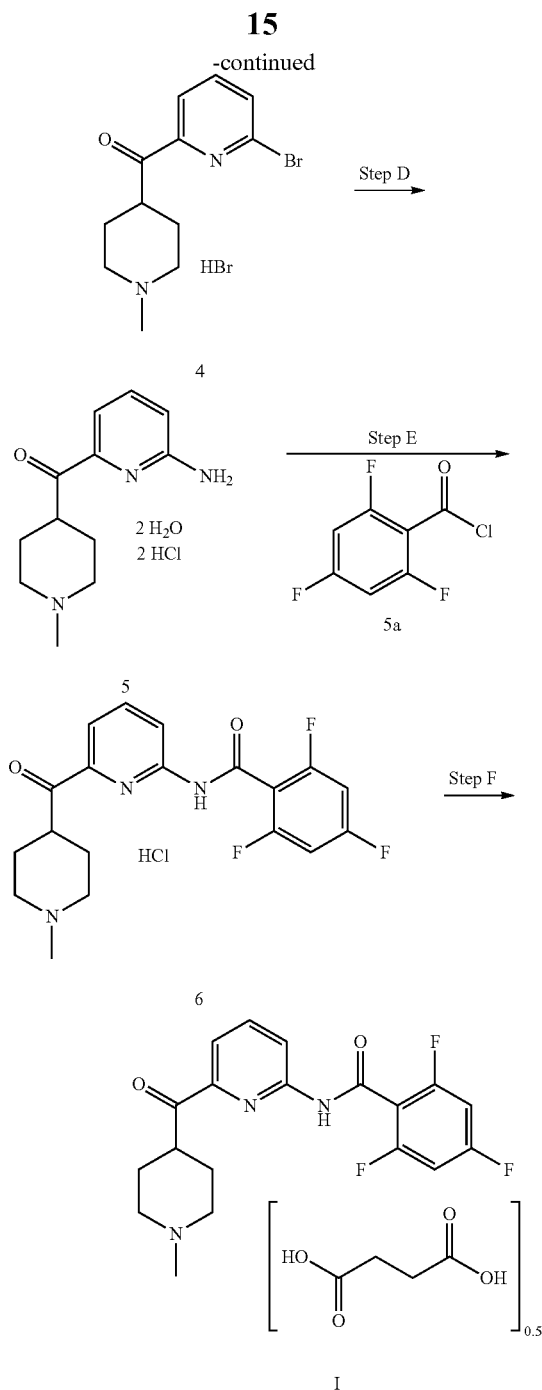
that compounds of the present invention are capable of forming salts. Some compounds of the present invention contain basic heterocycles, and accordingly react with any of a number of inorganic and organic acids to form pharmaceutically acceptable acid addition salts and common methodology for preparing them are well known in the art. See, e.g., P. Stahl, et al., HANDBOOK OF PHARMACEUTICAL SALTS: PROPERTIES, SELECTION AND USE, (VCHA/Wiley-VCH, 2008); S. M. Berge, et al., “Pharmaceutical Salts”, *Journal of Pharmaceutical Sciences*, Vol 66, No. 1, January 1977.

“Process scale” synthesis refers to preparations of 500 mg to 1000 kg, or more of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hemisuccinate. Preferably “process scale” syntheses are performed under Good Manufacturing Process (GMP) or similar conditions required for commercial production of pharmaceutical products for human consumption. Preferably, “process scale” in the processes of Route I and/or II above, refers to batches produced in at least 1 kilogram, and/or batches produced in at least 10 kilograms, and/or batches produced in at least 100 kilograms.

General Chemistry

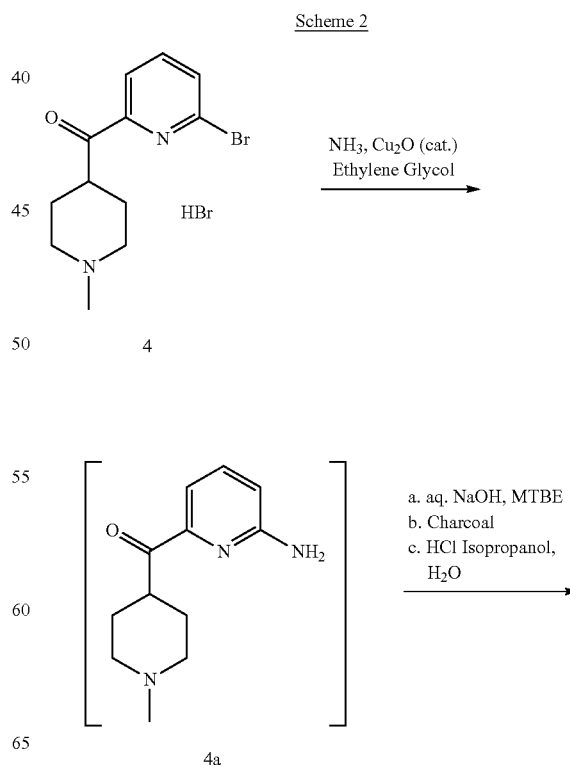
Scheme 1



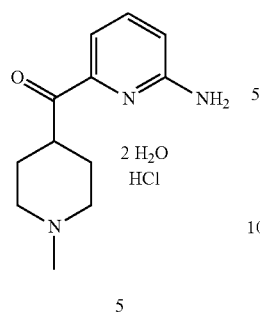


Scheme 1 depicts a process scale synthesis of lasmiditan hemisuccinate compound I. N-Methylation of commercially available piperidine 4-carboxylic acid 1 may be accomplished under various reductive conditions recognizable to the skilled artisan, specifically treatment of the secondary amine with about 1.3 equivalents of formaldehyde in an excess of formic acid, to obtain the N-methylpiperidine 2. Formation of diethylamide 3 may be achieved using conventional amide coupling reagents such as benzotriazole, HBTU or PyBOP or by converting the carboxylic acid to the acid chloride, using reagents well known in the art such as oxalyl chloride or thionyl chloride. More specifically, N-methylpiperidine-4-carboxylic acid 2 may be converted to the acid chloride by treatment with about 1.2 equivalents of thionyl chloride at about 50° C. for 1 hr, at which time the

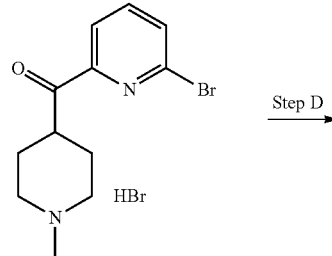
reaction mixture may be cooled to about 0° C. and 1.5 equivalents diethylamine and 3 equivalents trimethylamine added. The free base is stirred with HCl to obtain diethylamide hydrate hydrochloride 3. One skilled in the art will recognize that pyridyl ketone 4 may be obtained by treatment of diethylamide 3 with the lithiated bromopyridine 3a. More specifically, (6-bromo-2-pyridyl)lithium may be formed by treating 2,6 dibromopyridine with n-BuLi at about -58° C. Separately, piperidine-4-diethylamide hydrochloride hydrate 3 may be treated with about 2 equivalents NaOH and the resulting free base added to the lithiated species at about -58° C. The resulting mixture may be treated with HBr to form pyridylbromide hydrobromide 4. Amination of pyridylbromide hydrobromide 4 may be achieved using transition metal catalysis well known to one skilled in the art. More specifically, to pyridylbromide 4 may be added about 0.075 equivalents of Cu₂O, about 28 equivalents NH₃ in ethylene glycol and stirred to about 80° C. The reaction may be cooled to RT, quenched with H₂O, washed with 20% aqueous NaOH, slurried with 20% HCl in IPA and a small amount of H₂O, to obtain an aminopyridine dihydrate dihydrochloride 5 as a crystalline solid. Pyridylbenzamide hydrochloride 6 may be prepared by treating the free base of aminopyridyl 5 with the acid chloride 5a. More specifically, aminopyridine dihydrate dihydrochloride 5 may be treated with 6% aqueous NaOH to furnish the free base. Separately, 2,4,6-trifluorobenzoic acid may be treated with thionyl chloride at about 100° C. and the aforementioned freebase of 5, to provide pyridylbenzamide hydrochloride 6. Hemisuccinate I may be created by treating hydrochloride 6 with about 2 equivalents of NaHCO₃ followed by about 0.55 equivalents succinic acid to obtain lasmiditan hemisuccinate compound I.



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-continued



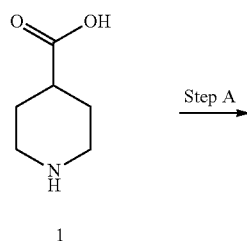
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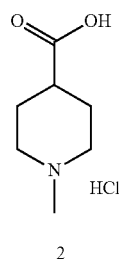
Step D

Scheme 2 depicts the synthesis of (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate hydrochloride 5. Amination of pyridylbromide hydrobromide 4 may be achieved as outlined in scheme 1 using transition metal catalysis well known to one skilled in the art. More specifically, to pyridylbromide 4 may be added about 0.075 equivalents of Cu_2O , about 28 equivalents NH_3 in ethylene glycol and stirred at about 80°C . The reaction may be cooled to RT, quenched with H_2O , washed with 20% aqueous NaOH , slurried with 20% HCl in IPA and a small amount of H_2O , to obtain aminopyridine dihydrate hydrochloride 5.

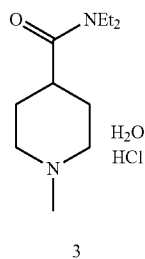
Scheme 3



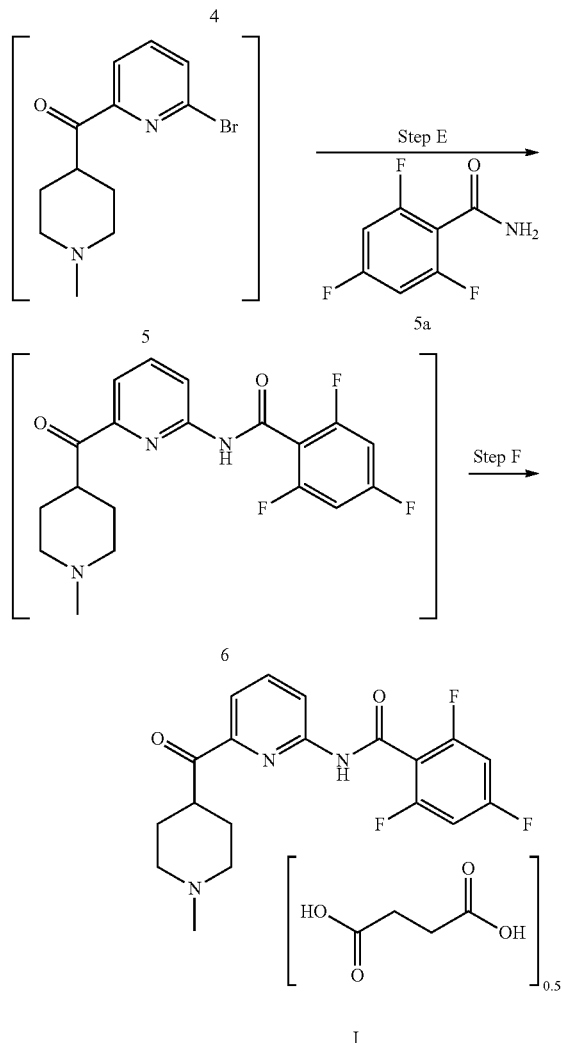
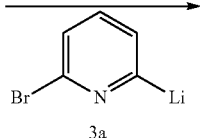
Step A



Step B



Step C



Step E

Step F

Scheme 3 illustrates a modified process synthesis to lasmiditan hemisuccinate I. N-Methylation of commercially available piperidine 4-carboxylic acid 1 may be accomplished under various reductive conditions recognizable to the skilled artisan, specifically treatment of the secondary amine with about 1.3 equivalents of formaldehyde in an excess of formic acid, to obtain the N-methylpiperidine 2. Formation of diethylamide 3 may be achieved using conventional amide coupling reagents such as benzotriazole, HBTU or PyBOP or by converting the carboxylic acid to the acid chloride, using reagents well known in the art such as oxalyl chloride or thionyl chloride. More specifically, N-methylpiperidine-4-carboxylic acid 2 may be converted to the acid chloride by treatment with about 1.2 equivalents

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of thionyl chloride at about 50° C. for 1 hr, at which time the reaction mixture may be cooled to about 0° C. and 1.5 equivalents diethylamine and 3 equivalents trimethylamine added. The free base is stirred with HCl to obtain diethylamide hydrate hydrochloride 3. One skilled in the art will recognize that pyridyl ketone 4 may be obtained by treatment of diethylamide 3 with the lithiated bromopyridine 3a. More specifically, (6-bromo-2-pyridyl)lithium may be formed by treating 2,6 dibromopyridine with n-BuLi at about -58° C. Separately, piperidine-4-diethylamide hydrochloride hydrate 3 may be treated with about 2 equivalents NaOH and the resulting free base added to the lithiated species at about -58° C. The resulting mixture may be treated with HBr to form pyridylbromide hydrobromide 4. Amination of pyridylbromide hydrobromide 4 to obtain amide 6 may be achieved using transition metal catalysis well known to one skilled in the art. Specifically, the pyridyl ketone 4 may be sprung to its corresponding free base form with a suitable mineral base and subjected to Buchwald-type coupling conditions, as is well known in the literature. More specifically, the free base of compound 4 may be stirred in a suitable aprotic solvent, such as toluene or xylene, containing a mixture of about 1-5 weight % water, about 1.1 equivalents commercially available 2,4,6-trifluorobenzamide (CAS #82019-50-9), about 1.5 equivalents of potassium carbonate, about 0.005 to about 0.015 equivalents of a suitable palladium catalyst, such as palladium(II) acetate, and about 0.01 to 0.02 equivalents of a suitable phosphine ligand compound, such as Xantphos, XPhos, or DPEPhos. The resulting mixture may be heated at about 70° C. for about 12-24 hr. The reaction mixture may be diluted with a suitable mixture of water and organic solvent, such as DCM or EtOAc, and the organic layer may be treated with an appropriate palladium scavenger, such as thiourea-modified silica gel, for about 8-24 hr at about RT to about 65° C. The resulting mixture may be cooled, filtered, treated with activated charcoal, filtered, and concentrated under reduced pressure. The resulting residue may be dissolved in an appropriate alcoholic solvent, such as ethanol, and treated slowly with a solution of about 0.5 equivalents of succinic acid dissolved in ethanol at about 55° C. The resulting mixture may be cooled to RT over about 10 hr, and the resulting slurry may be slurry-milled by treatment under a series of thermal cycles of heating to 60° C. and cooling back to RT over 4 hr. The resulting solid may be collected by filtration, dried at about 40° C. for about 4 hr, and optionally jet milled, to obtain lasmiditan hemisuccinate I.

Experimental Procedures

The following preparations of process intermediates further illustrate the invention and represent typical syntheses of various compounds. The reagents and starting materials are readily available or may be readily synthesized by one of ordinary skill in the art. It should be understood that the Preparations and Examples are set forth by way of illustration, and that various modifications may be made by one of ordinary skill in the art.

LC-ES/MS is performed on an AGILENT® HP1100 liquid chromatography system. Electrospray mass spectrometry measurements (acquired in positive and/or negative mode) are performed on a Mass Selective Detector quadrupole mass spectrometer interfaced to the HP1100 HPLC. LC-MS conditions (low pH): column: PHENOMENEX® GEMINI® NX C18 2.1 mm×50 mm, 3.0µ; gradient: 5-100% B in 3 min, then 100% B for 0.75 min column temperature: 50° C.±10° C.; flow rate: 1.2 mL/min; Solvent A: deion-

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ized water with 0.1% HCOOH; Solvent B: ACN with 0.1% formic acid; wavelength 214 nm. Alternate LC-MS conditions (high pH): column: XTERRA® MS C18 columns 2.1×50 mm, 3.5 µm; gradient: 5% of solvent A for 0.25 min, gradient from 5% to 100% of solvent B in 3 min and 100% of solvent B for 0.5 min or 10% to 100% of solvent B in 3 min and at 100% of solvent B for 0.75 min; column temperature: 50° C.±10° C.; flow rate: 1.2 mL/min; Solvent A: 10 mM NH₄HCO₃ pH 9; Solvent B: ACN; wavelength: 214 nm.

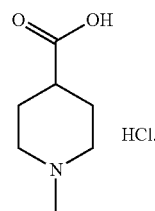
NMR spectra are performed on a Bruker A VIII HD 400 or 500 MHz NMR Spectrometer, obtained as CDCl₃ or (CD₃)₂SO solutions reported in ppm, using residual solvent [CDCl₃, 7.26 ppm; (CD₃)₂SO, 2.05 ppm] as reference standard. When peak multiplicities are reported, the following abbreviations may be used: s (singlet), d (doublet), t (triplet), q (quartet), m (multiplet), br-s (broad singlet), dd (doublet of doublets), dt (doublet of triplets). Coupling constants (J), when reported, are reported in hertz (Hz).

Chloride analysis is performed on an ESA CORONA® Plus instrument equipped with a CORONA® CAD® (charged aerosol detector)-HPLC, Acclaim Trinity P1 (100×3.0 mm, 3 µm), mobile phase: 50 mM ammonium acetate, pH ~5 in ACN.

The compounds described herein can be prepared by general methods known to the skilled artisan or by processes described herein. Suitable reaction conditions for the steps of these schemes are well known in the art and appropriate substitutions of solvents and co-reagents are within the skill of the art. Likewise, it will be appreciated by those skilled in the art that synthetic intermediates may be isolated and/or purified by various well-known techniques as needed or desired, and that frequently, it will be possible to use various intermediates directly in subsequent synthetic steps with little or no purification. Furthermore, the skilled artisan will appreciate that in some circumstances, the order in which moieties are introduced is not critical.

Preparation 1

1-methylpiperidine-4-carboxylic acid hydrochloride



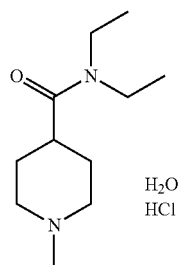
Scheme 1, step A: To a jacketed reactor is charged piperidine-4-carboxylic acid (10.0 g, 77.5 mmol) and deionized water (40 mL). The mixture is heated to reflux (95-100° C.). Formic acid (13.9 g, 302 mmol) is added over 30 min. A 37% aqueous solution of formaldehyde (8.1 g, 101 mmol) is added to the mixture dropwise over at least 30 min. Water (0.3 mL) is used as a line rinse into the reactor. The mixture is stirred for 4 hr at reflux (95-100° C.) and sampled by HPLC for IPC analysis (NMT 0.5% of piperidine-4-carboxylic acid). If the amount of piperidine-4-carboxylic acid is above 0.5%, the mixture is stirred 2 additional hr. If the specification is met, the solution is concentrated under vacuum until ~20 mL of residual volume remains and the

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residue is cooled to 45-50° C. To the cooled solution is charged 33% aqueous HCl (12.8 g, 116 mmol) over not less than 30 min. Water (0.3 mL) is used as a line rinse into the reactor. Water is distilled off under vacuum until ~20 mL of residual volume remains. To the concentrated solution at 45-50° C. is charged ACN (42.4 mL) and the mixture is concentrated under atmospheric pressure until ~40 mL of residual volume remains. To the concentrated solution at 45-50° C. is charged ACN (20.4 mL) and the mixture is concentrated under atmospheric pressure until ~40 mL of residual volume remains. The dilution/concentration operations are repeated until the in process control for water content by Karl-Fischer analysis is NMT 0.2%; during these operations a slurry forms. To the slurry is charged ACN (10.2 mL) at 45-50° C. The slurry is cooled to 20° C. over 1 h and stirred for an additional 2 h. The resulting solid is isolated by filtration and the cake is rinsed with ACN (10.2 mL). The wet cake is dried at 40° C. under nitrogen at atmospheric pressure to give the title compound (12.1 g, 87% yield). MS (m/z): 144 (M+H).

Preparation 2

N,N-diethyl-1-methyl-piperidine-4-carboxamide hydrate hydrochloride



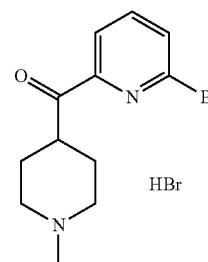
Scheme 1, step B: To a jacketed reactor is charged 1-methylpiperidine-4-carboxylic acid hydrochloride (30.0 g, 167 mmol), chlorobenzene (240 mL) and DMF (0.61 g, 8.35 mmol) and the resulting mixture is heated to 50° C. To the hot suspension is charged thionyl chloride (24.2 g, 200.4 mmol) over a 1 hr period. Chlorobenzene (13.5 mL) is used as a line rinse into the reactor. The mixture is stirred for 5 hr after the completion of the thionyl chloride addition. The solution is then cooled to 0 to 10° C. A solution prepared from diethylamine (17.7 g, 12.5 mmol) and TEA (50.7 g, 25 mmol) is charged to the cold reaction mixture over a 3 hr period. Chlorobenzene (13.5 mL) is used as a line rinse into the reactor. The mixture is stirred for 2 hr after the complete addition of the amine mixture. The reaction is treated with 20 weight % aqueous NaOH (180.3 g, 902 mmol) and stirred at RT for 2 hr. Water (3 mL) is used as a line rinse into the reactor. The mixture is allowed to settle for 2 hr and the aqueous phase is removed. The remaining organic phase is placed under vacuum. The mixture is heated to distill away the residual amines as well as most of the chlorobenzene. The reactor is vented to atmospheric pressure using nitrogen after approximately ten volumes of distillate have been collected. The remaining solution is cooled to between 10° C. to 30° C. THF (120 mL) and water (4.54 g, 252 mmol) are charged to the reactor. With the reaction mixture at RT, the desired product is precipitated by the addition of 20 weight % aqueous HCl in isopropanol (30.4 g, 167 mmol).

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THF (5.4 mL) is used as a line rinse into the reactor. After the complete addition of HCl, the suspension is stirred for 2 hr at RT. The resulting solid is collected by filtration and washed with THF (75.0 mL). The collected solids are dried under vacuum for 16 hr at 40° C. to give the title compound (35.5 g, 84% yield). MS (m/z): 199 (M+H).

Preparation 3

(6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone hydrobromide



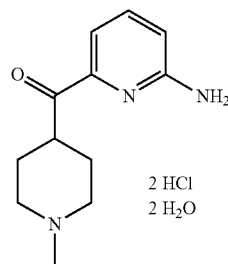
Scheme 1, step C: A suspension of N,N-diethyl-1-methyl-piperidine-4-carboxamide hydrate hydrochloride (21.5 g, 85.1 mmol) in MTBE (109 mL) is treated with a 20 weight % aqueous solution of NaOH (34.0 g, 170 mmol). A water rinse (1.94 mL) is used to complete the addition. The mixture is stirred at RT for 30 min, the phases are allowed to settle, and phases are separated. The aqueous phase is extracted with MTBE (43.7 mL) and the organic phases combined. The organic phase is dried by distillation at atmospheric pressure until the in process control for water content by Karl-Fischer analysis is <0.10 weight %. If the target analysis is not met, the reaction is charged with MTBE (43.7 mL) and the distillation is repeated. Typically three distillations are required to reach the target analysis for water. In a separate reactor is charged a mixture of 2,6-dibromopyridine (30.2 g, 128 mmol) and MTBE (105 mL) and is cooled to less than -58° C. To the cooled suspension is charged a 2.5M solution of n-BuLi in hexanes (51.3 mL, 128 mmol) over a 2 hr period. A rinse of MTBE (4.5 mL) is used to complete the transfer. The mixture is aged while maintaining the temperature at less than -58° C. for an additional 2 hr after complete n-BuLi addition. After aging, the solution of N,N-diethyl-1-methyl-piperidine-4-carboxamide hydrate hydrochloride in MTBE is added to the cold reaction over a 45 min period. A rinse of MTBE (13.5 mL) is used to complete the transfer. The mixture is aged for at least 30 min after complete addition of N,N-diethyl-1-methyl-piperidine-4-carboxamide hydrate hydrochloride in MTBE. After aging, the reaction is warmed to 0° C. over 1 hr. The cold reaction mixture is added to a 2.5 M aqueous solution of HCl (146 mL, 366 mmol) at a rate to maintain the quench temperature at NMT 30° C. A rinse of MTBE (13.5 mL) is used to complete the transfer. The mixture is stirred for at least 30 min after the transfer is complete and the phases are allowed to settle. The phases are separated and the aqueous phase is retained. n-BuOH (54.8 mL) is added to the aqueous phase and the mixture is treated with a 20 weight % aqueous solution of NaOH (59.5 g, 298 mmol). A rinse of water (2.80 mL) is used to complete the transfer. The mixture is stirred for at least 30 min and the phases are allowed to settle. The phases are separated and the organic

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phase is retained. The aqueous phase is extracted with n-BuOH (54.8 mL). The combined organic phases are dried by distillation under vacuum to obtain an in process control for water content by Karl-Fischer analysis of <0.20 weight %. If the target analysis is not met, n-BuOH (41.1 mL) is charged and the distillation is repeated. Typically, two distillations are required to reach the in process control target analysis. The concentrated solution is clarified by filtration and a rinse with n-BuOH (89.6 mL) is used to complete the transfer and rinse the filter. The clarified solution is treated with a 48 weight % aqueous solution of HBr (9.91 mL, 87.7 mmol) over a 90 min period. A rinse of n-butanol (13.8 mL) is used to complete the transfer. A check of the pH shows the reaction mixture has a pH ~1. The mixture is dried by distillation at atmospheric pressure to obtain an in process control for water content by Karl-Fischer analysis of <0.30 weight %. The mixture is concentrated to 172 mL. If the target analysis is not met, n-BuOH (54.8 mL) is charged and the distillation is repeated. The mixture is cooled to 20° C. and stirred for 12 hr. The resulting solids are collected by filtration and washed twice with n-BuOH (10.75 mL). The solids are dried under vacuum at 60° C. to obtain the title compound (24.8 g, 80% yield). MS (m/z): 283, 285 (⁷⁹Br, ⁸¹Br, M+H).

Preparation 4

(6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride



Scheme 1, step D: To a pressure reactor is charged (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone hydrobromide (30 g, 82.9 mmol) and Cu₂O (880 mg, 6.2 mmol). The headspace is exchanged with nitrogen/vacuum purge cycles three times. To the solids are charged a solution of NH₃/ethylene glycol (273.5 g total, 39.1 g NH₃, 2.33 mol; 210 mL ethylene glycol) and the resulting mixture is stirred at RT for 2 hr. The mixture is heated to 80° C., stirred for 10 h, and cooled to RT for in process control sampling for (6-bromo-2-pyridyl)-(1-methyl-4-piperidyl)methanone hydrobromide NMT 2%. If the target analysis is not met, the reaction is stirred for another 4 hr at 80° C. and sampled again. To the completed reaction is charged H₂O (90 mL) and the mixture is filtered. The filtrate is charged into aqueous NaCl (253.9 g NaCl, 2.73 mol, 13.7 L/kg H₂O) and the resulting mixture is stirred at RT for 10 min. To the mixture is charged a 20% aqueous solution of NaOH (4.44 equiv., 368 mmol) and the biphasic mixture is stirred at RT for 5 min. The mixture is extracted with MTBE (90 mL) 4-5 times at RT. The combined MTBE layers are treated with 5 weight % carbon for 30 min and the carbon is removed by filtration. The organic filtrate is concentrated to ~150 mL under vacuum. To the concentrated filtrate is charged IPA

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(200 mL) and the solution is concentrated to ~150 mL under vacuum two times. The IPA distillations are repeated as needed to meet the target analyses for in process control of water. Water content is confirmed to be not more than 0.2% by Karl-Fischer analysis. In a separate reactor is charged a 20 weight % solution of HCl in IPA (30 g, 166 mmol) and water (10.5 mL) at RT. The concentrated product mixture is charged to the HCl solution over 90 min. The resultant slurry is stirred at RT for not less than 8 hr. The slurry is filtered, rinsed twice with a mixture of 95:5 IPA/H₂O (36 mL) at RT, and dried under vacuum at 40° C. for 16-24 hr to afford the title compound (18.4 g 68% yield). MS (m/z): 220 (M+H). ¹H NMR (400 MHz, D₂O/DMSO-d₆) δ ppm 1.74-1.88 (m, 2H), 2.05 (br d, J=14.9 Hz, 2H), 2.73 (s, 3H), 3.01 (td, J=13.1, 2.6 Hz, 2H), 3.41-3.50 (m, 2H), 3.55 (tt, J=12.0, 3.5 Hz, 1H), 7.14 (dd, J=9.0, 0.7 Hz, 1H), 7.59 (d, J=7.2 Hz, 1H), 7.90 (dd, J=9.0, 7.2 Hz, 1H). ¹³C NMR (101 MHz, D₂O/DMSO-d₆) δ ppm 27.4, 40.7, 44.6, 54.6, 117.2, 121.1, 137.9, 145.2, 156.2, 196.3. Chloride analysis: 20.23% (n=2).

X-Ray Powder Diffraction (XRPD) of Crystalline Forms

The XRPD patterns of crystalline solids are obtained on a Bruker D4 Endeavor X-ray powder diffractometer, equipped with a CuKα source and a Vantec detector, operating at 35 kV and 50 mA. The sample is scanned between 4 and 40 2θ°, with a step size of 0.008 2θ° and a scan rate of 0.5 seconds/step, and using 1.0 mm divergence, 6.6 mm fixed anti-scatter, and 11.3 mm detector slits. The dry powder is packed on a quartz sample holder and a smooth surface is obtained using a glass slide. The crystal form diffraction patterns are collected at ambient temperature and relative humidity. Crystal peak positions are determined in MDI-Jade after whole pattern shifting based on an internal NIST 675 standard with peaks at 8.853 and 26.774 2θ°. It is well known in the crystallography art that, for any given crystal form, the relative intensities of the diffraction peaks may vary due to preferred orientation resulting from factors such as crystal morphology and habit. Where the effects of preferred orientation are present, peak intensities are altered, but the characteristic peak positions of the polymorph are unchanged. See, e.g. The United States Pharmacopeia #23, National Formulary #18, pages 1843-1844, 1995. Furthermore, it is also well known in the crystallography art that for any given crystal form the angular peak positions may vary slightly. For example, peak positions can shift due to a variation in the temperature at which a sample is analyzed, sample displacement, or the presence or absence of an internal standard. In the present case, a peak position variability of ±0.2 2θ° is presumed to take into account these potential variations without hindering the unequivocal identification of the indicated crystal form. Confirmation of a crystal form may be made based on any unique combination of distinguishing peaks.

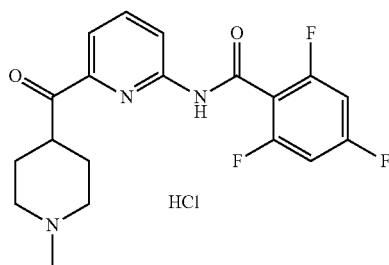
A sample of Preparation 4, (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride, is characterized by an XRD pattern using CuKα radiation as having diffraction peaks (2θ values) as described in Table 1 below, and in particular having peaks at 8.3° in combination with one or more of the peaks selected from the group consisting of 16.6°, 23.5°, and 33.7°, with a tolerance for the diffraction angles of 0.2 degrees.

TABLE 1

X-ray powder diffraction peaks of the crystalline compound of Preparation 4; (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate dihydrochloride		
Peak	Angle (° 2θ) +/- 0.2°	Relative Intensity (% of most intense peak)
1	8.3	100.0%
2	16.6	49.8%
3	19.9	8.1%
4	22.5	15.2%
5	23.5	25.7%
6	25.1	17.1%
7	28.8	11.0%
8	29.7	17.0%
9	30.0	13.9%
10	33.7	23.7%

Preparation 5

2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hydrochloride



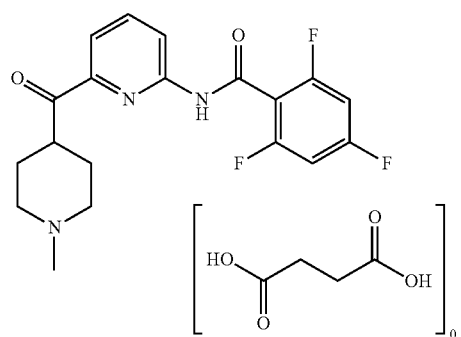
Scheme 1, step E: To a suspension of (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrate (10 g, 30.6 mmol) in chlorobenzene (65 mL) is charged 6 w/w % aqueous NaOH (3 g, 75 mmol). The biphasic mixture is heated to 54° C. with stirring for 30 min, the mixture is allowed to separate over 30 min, and the layers are separated at 54° C. The aqueous layer is back-extracted with chlorobenzene (45 mL) at RT. The organic layers are combined and distilled under vacuum to ~62 mL to afford a solution of (6-aminopyridin-2-yl)(1-methylpiperidin-4-yl)methanone. In a separate reactor is charged 2,4,6-trifluorobenzoic acid (5.9 g, 1.1 eq, 33.7 mmol), DMF (62 mg, 0.85 mmol) and chlorobenzene (32 mL) and the mixture is heated to 80° C. To the heated mixture is charged thionyl chloride (4.37 g, 37 mmol) over 4 hr at 80° C. The mixture is stirred at 80° C. for at least 6 hr, and heated to 100° C. for at least 6 hr to purge residual HCl gas. The solution of acid chloride is cooled to RT and transferred to a separate reactor. The acid chloride solution is heated to 100° C. and charged with (6-aminopyridin-2-yl)(1-methylpiperidin-4-yl)methanone over 4 hr. The resulting slurry is agitated for an additional 3 hr at 100° C. and cooled to RT. To the cooled slurry is charged ACN (100 mL). The resulting slurry is heated to 80° C. for 1 hr and cooled to RT over 2 hr. The resulting slurry is further agitated at RT for an additional 1 hr and filtered. The filter cake is washed with ACN (10 mL) at RT. The collected solids are dried under vacuum at 100° C. for 16 hr to obtain the title compound (10.7 g, 85% yield). MS m/z 378 (M+H).

Alternate Procedure For Preparation 5

To a suspension of (6-amino-2-pyridyl)-(1-methyl-4-piperidyl)methanone dihydrochloride dihydrate (10 g, 30.6 mmol) in chlorobenzene (65 mL) is charged 6 w/w % aqueous NaOH (2.97 g, 74.4 mmol). The biphasic mixture is heated to 54° C. with agitation for 30 minutes and the layers are allowed to separate over 30 min. The layers are separated at 54° C. The aqueous layer is back-extracted with chlorobenzene (45 mL) at RT. The organic layers are combined and distilled under vacuum to ~62 mL to afford a solution of (6-aminopyridin-2-yl)(1-methylpiperidin-4-yl)methanone. In a separate reactor is charged 2,4,6-trifluorobenzoic acid (5.9 g, 33.7 mmol), DMF (62 mg, 0.85 mmol) and chlorobenzene (32 mL) and the mixture is heated to 80° C. To the heated mixture is charged thionyl chloride (4.4 g, 37 mmol) over 4 hr at 80° C. The mixture is agitated at 80° C. for at least 6 hours and heated to 100° C. for at least 6 hours to purge residual HCl gas. The solution of acid chloride is cooled to RT and transferred to a separate reactor. The acid chloride solution is heated to 100° C. and to the solution is charged (6-aminopyridin-2-yl)(1-methylpiperidin-4-yl)methanone over 4 hr. The resulting slurry is agitated for an additional 3 hours at 100° C. and cooled to RT. To the cooled slurry is charged ACN (100 mL). The resulting slurry is heated to 80° C. for 1 hr and cooled to RT over 2 hr. The resulting slurry is further agitated at RT for an additional 1 hr and the resulting solids are collected by filtration. The filter cake is washed with ACN (10 mL) at RT. The solids are dried under vacuum at 100° C. for 16 hr to obtain the title compound (10.7 g, 85% yield). MS m/z 378 (M+H).

Preparation 6

2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hemisuccinate



Scheme 1, step F: To a reactor is charged 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide hydrochloride (20 g, 48.4 mmol) and MTBE (202 mL). To the stirred slurry at RT is charged a solution of aqueous NaHCO₃ (8.13 g, 96.8 mmol NaHCO₃ in 200 mL water) over 1 hr. The biphasic mixture is separated and the aqueous layer is back-extracted with MTBE (202 mL). The combined organic layers are distilled under vacuum to a final volume of ~200 mL. To the distilled solution is charged SiO₂ (2 g), the resulting mixture is stirred for 30 min at RT, filtered, and the filter cake is rinsed with MTBE (10.8 mL). To the filtrate is charged carbon (340 mg; alternatively, the solution can be filtered through a carbon cartridge) and the resulting mixture

TABLE 2-continued

Unit Formula for Lasmiditan 50 mg and 100 mg Tablets			
Component	Quantity		Description
	50 mg	100 mg	
Microcrystalline Cellulose ^a	30.86	61.71	Filler
Pregelatinized Starch	7.500	15.00	Binder
Croscarmellose Sodium	5.630	11.26	Disintegrant
Sodium Lauryl Sulfate,	0.5600	1.120	Wetting Agent
Purified Water ^b	—	—	Granulating Liquid
	(4.774) ^c	(9.548) ^c	
Core Tablet - Extragranular Components			
Croscarmellose Sodium	7.880	15.76	Disintegrant
Magnesium Stearate	2.250	4.500	Lubricant
Core Tablet Weight:	117.3	234.5	
Film Coating			
Color Mixture Gray	3.519	—	Colorant
Color Mixture Purple	—	7.035	Colorant
Purified Water ^d	—	—	Solvent
Total Tablet Weight:	120.8	241.5	

Table 2 Notes:

^aA salt conversion factor of 0.86469 is used to calculate the quantity of lasmiditan hemisuccinate. The quantity of microcrystalline cellulose may be adjusted accordingly to maintain target tablet weight.

^bPurified Water is used in the granulation operation. The majority of the water is subsequently removed during the drying operation.

^cA small quantity of residual water remains following the drying process, which may be in the form of free water or as water of hydration associated with drug substance.

^dPurified water is used in the coating unit operation. The coating suspension is comprised of 20% w/w solids. Sufficient coating is sprayed to target a weight gain of 3%. This water is removed during the coating unit operation.

Tablet Manufacture:

Lasmiditan tablets are manufactured using a high shear wet granulation process which is described as follows. High Shear Wet Granulation: Sodium lauryl sulfate is passed through a security screen and added to purified water to form the granulating liquid. Lasmiditan drug substance and the excipients to be wet granulated (microcrystalline cellulose, pregelatinized starch, croscarmellose sodium) are passed through a security screen and combined in the granulator. The materials are mixed with the main impeller of the granulator prior to the addition of the granulating liquid. The powder blend is granulated in the granulator by adding the granulating liquid, while the powder is mixing. Upon completion of the liquid addition, the granulation is wet massed to facilitate liquid distribution. The granulation is coarsely sized by passing through a cone mill prior to drying.

Fluidized Bed Drying: The granulation is dried in a fluidized bed dryer until a moisture value of (50 mg and 100 mg: NMT 7%) is achieved, as measured by a gravimetric loss on drying method, or using a scientifically justified equivalent method. The dried granules are passed through a cone mill and added to a tumble bin.

Final Blend—Extragranular Powder Blend and Final Blend Lubrication: The extragranular croscarmellose sodium is passed through a security screen, and added to the dry milled granules in the tumble bin. The materials are tumble blended. The extragranular magnesium stearate is passed through a security screen, and added to the tumble bin. The materials are tumble blended.

Tablet Compaction: The blended granulation is compressed into tablets using a rotary compression machine. The compression parameters are selected at the start of a batch to achieve the target average (n=10) tablet weight of (50 mg: 111.4 mg-123.2 mg, 100 mg: 222.8-246.2 mg) and average (n=10) tablet breaking force of (50 mg: 4.1-13.7 kiloponds, 100 mg: 6.0-17.9 kiloponds). The average (n=10) core tablet

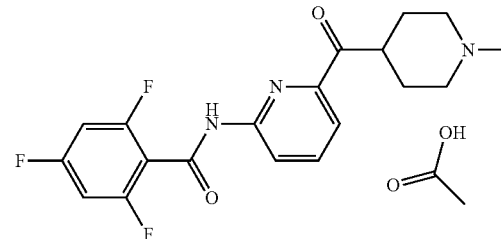
weight, tablet breaking force, and tablet thickness are evaluated during start-up and throughout the compression unit operation. Tablet friability and tablet disintegration time are evaluated at the start-up of compression.

5 Film Coating of Core Tablets: The color mixture (gray for the 50 mg, and purple for the 100 mg) is passed through a security screen and mixed with purified water to form the coating suspension. The tablets are film-coated with the suspension utilizing spray guns in a perforated coating pan. 10 The pan is rotated while the coating suspension is applied at a controlled rate with pneumatic atomization, and drying air is passed through the tablet bed to yield an acceptable exhaust temperature. Sufficient coating is sprayed to achieve the desired percent coating applied (50 mg and 100 mg: 2.0%-5.5%). The film-coated tablets are inspected for visual quality following the completion of coating step. The film-coated tablets are discharged into bulk storage containers and may be sorted (optional).

20 Container Closure System—Unit-Dose Blisters:

Lasmiditan tablets are provided in individual blister cavities formed from polychlorotrifluoroethylene (PCTFE)/polyvinylchloride (PVC) laminated film and sealed with aluminum foil laminate lidding material which contains a PVC-based heat seal coating.

Preparation and Description of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide acetate



Preparation of 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide acetate (also referred to as lasmiditan acetate) is performed by placing 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide (541 mg, 1.6 mmol) in isopropyl acetate (5 mL) while stirring at 1000 rpm at room temperature. Acetic acid (100 35 μL) is added. A white solid precipitates out of solution after about two minutes of stirring. Stirring is shut off after 10 minutes, the white solid is collected by vacuum filtration on Whatman paper and dried in place under air stream for 10 minutes to yield the title compound (650 mg, 92% yield).

Counterion stoichiometry is measured by nuclear magnetic resonance using an Agilent 400-MHz spectrometer. A sample solution is prepared by dissolving the prepared 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide acetate (7.715 mg) and maleic acid (5.949 mg), used as standard for other measurements, in 60 DMSO-d₆ (approximately 0.75 mL). A ¹³C-decoupled ¹H spectrum of the sample from 0-12 ppm is acquired using the following parameters: 90-degree excitation pulse, 64 scans, 25-second relaxation delay, and 4.5-second acquisition time. The resonances for acetate at approximately 1.9 ppm (3H) and for 2,4,6-trifluoro-N-[6-(1-methylpiperidine-4-carbonyl)-2-pyridyl]benzamide at approximately 7.8 ppm (1H) are integrated to obtain areas of 29094 and 9508, respec-